

- 5 (a)** The work function of a surface is the minimum energy of the individual photons in the incident radiation on the surface required to cause photoemission of electrons.

Note: Must *mention minimum* energy for work function!

- (b)** The energy of the photons

$$E = \frac{hc}{\lambda} = (6.63 \times 10^{-34})(3 \times 10^8) / (540 \times 10^{-9})$$
$$= 3.683 \times 10^{-19} \text{ J} = 2.3 \text{ eV}$$

Since the energy of the incident photon is lower than work function of metal, no electrons are emitted from surface.

- (c)** Intensity only increases number of photons incident on metal surface per unit time per unit area. If wavelength remains unchanged and photon energy is still lower than work function, no photoelectrons emitted.

Note: Rate of arrival of photons is critical in the explanation.

- 6 (a) (i)** The resultant force acting on a body is the rate of change of momentum of the